D06 · Carrier Management & TMS

02 · TMS implementation & transportation planning

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1 WHAT IT IS

A Transportation Management System (TMS) plans, executes and settles freight: it turns orders into loads, selects and books carriers, tracks shipments and audits invoices, integrated with the ERP and WMS.

2 WHY IT MATTERS

Without a TMS, transport planning lives in spreadsheets and phone calls: loads are built by habit, carriers are chosen by relationship, and invoices are paid without checking. **The TMS turns those decisions into rules that can be optimized and audited.** But a TMS automates whatever process it receives — poor master data and undefined rules produce faster chaos, not better freight.

3 HOW IT IS DONE

1. **Define the process and rules first.** Routing guide, carrier selection logic, tender rules and exception handling, documented before configuration.
2. **Clean the master data.** Locations, lanes, rates, carriers and product dimensions — the optimizer is only as good as its inputs.
3. **Integrate ERP, WMS and TMS.** Orders flow in, shipment confirmations and freight costs flow back; no re-keying.
4. **Phase the rollout.** Booking and tendering first, then load optimization, then freight audit and settlement — measure benefits at each step.

IN PRACTICE

ERP–TMS integration is non-negotiable

Without it the TMS becomes another disconnected system that creates work instead of removing it.

Define optimization parameters before configuration

Changing parameters during implementation is the most frequent cause of delays and cost overruns.

Treat change management as 50% of the project

A TMS that is used only as a recording system instead of an optimization engine delivers a fraction of its potential value.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — FMCG company, 1,200 transport orders/month — Logistics team spent ~70% of time on manual coordination.

Module	Timeline	Impact
Order Management + auto booking	3 months	Booking time 45 min -> 8 min (-82%)
Load Planning + route optimization	4 months	Truck fill rate 74% -> 88% · Freight -9%
Real-time tracking	4 months	In-transit visibility 0% -> 94%
Automated freight audit	5 months	$84K incorrect charges recovered in first 6 months

Result: Freight cost -11.4% ($480K/year). Team capacity freed allowed reallocation of 3 FTEs to network optimization projects.

HOW IT IS MEASURED

TMS Automation Rate %

(Orders processed without manual intervention / Total orders) × 100

Benchmark: >80% of orders processed automatically

TMS-Driven Savings %

(Post-TMS freight cost - Pre-TMS) / Pre-TMS × 100

Benchmark: 8–15% freight reduction in the first 12 months is typical for a well-executed implementation

WHAT YOU WOULD USE

TMS Platforms

SAP TM — Native integration with S/4HANA, strong for complex multi-modal environments.

Blue Yonder TMS / Oracle OTM / Manhattan — Best-of-breed options with advanced optimization engines.

4 SO WHAT

Fix the process, then automate it. Document the rules, clean the data, integrate with ERP and WMS, and roll out in phases that each pay for themselves.

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